

Boundary-Aware Agricultural Field Segmentation

Comparing auxiliary distance, direct weighting, and explicit boundary prediction

HLCV Team 004

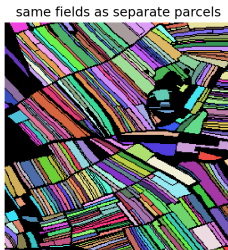
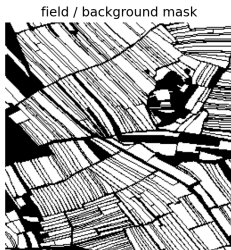
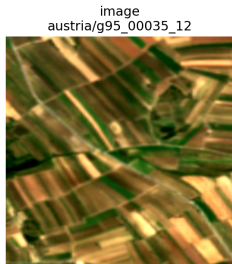
Saarland University

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Four controlled methods

1,430 held-out test chips

The challenge is preserving individual fields, not only field area



Failure mode

A binary mask can cover the correct area while merging neighboring fields.

Project goal

Improve border agreement without sacrificing region overlap.

Source: FTW imagery and labels; visualization generated in this project.

Data and supervision

Fields of The World subset

Countries	Austria, Croatia, Denmark
Input	two seasonal images, each with RGB+NIR (8 channels total)
Resolution	256 × 256 pixels; 10 m per pixel
Labels	field mask + FTW instance mask

10,938 train

1,347 validation

1,430 test

How the labels are used

Binary field mask

Mask head

Direct target: field or background at every pixel.

FTW instance mask

Provided by the dataset; separates neighboring fields.

converted into ↓

Signed distance field (SDF)

Distance head

Derived target: distance to the nearest field boundary.

Source: Kerner et al., "Fields of The World," AAAI 2025.

Related work and what we contributed

Built upon

- ▶ **FTW**: global, multi-date Sentinel-2 field benchmark.
- ▶ **U-Net**: encoder–decoder with skip connections for dense prediction.
- ▶ **Distance-aware supervision**: geometry can complement region losses.

Our project work

- ▶ Added an instance-derived SDF target and a second regression head.
- ▶ Added an SDF-based weighting map to focus BCE near field borders.
- ▶ Added an explicit boundary-prediction head and compared all checkpoints on the test set.

Research question: where should boundary information enter the learning objective?

Source: Ronneberger et al., MICCAI 2015; Kervadec et al., MIDL 2019 / MedIA 2021.

Four methods, one controlled comparison

Method	Architecture	Training loss	Boundary signal
Mask baseline	one mask head	$\mathcal{L}_{BCE} + \mathcal{L}_{Dice}$	none
Two-head SDF	mask + distance heads	$\mathcal{L}_{BCE} + \mathcal{L}_{Dice} + 0.1\mathcal{L}_{SDF}$	auxiliary geometry
Boundary BCE	one mask head	$\mathcal{L}_{WBCE} + \mathcal{L}_{Dice}$	direct mask gradients
Mask + boundary	two active heads	$\mathcal{L}_{WBCE} + \mathcal{L}_{Dice} + 2\mathcal{L}_{Bnd}$	mask gradients + explicit border task

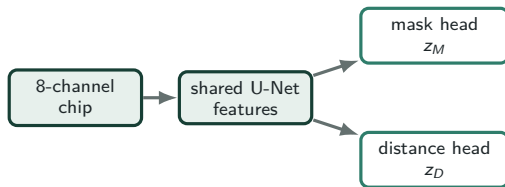
Held constant

Data, augmentation, U-Net width, optimizer, and evaluation code.

Checkpoint audit

The trained mask baseline uses ordinary **BCE + Dice**.

Method 1: a second head predicts signed distance



Hypothesis

The SDF task forces shared features to encode individual field geometry.

- ▶ $D > 0$: inside an individual field
- ▶ $D \approx 0$: field boundary
- ▶ $D < 0$: background
- ▶ Inference still uses the mask head

+9,313 parameters

+0.12%

Joint objective

$$\mathcal{L}_{2H} = \mathcal{L}_{BCE}(z_M, Y) + \mathcal{L}_{Dice}(\sigma(z_M), Y) \\ + 0.1 \text{SmoothL1}(\tanh(z_D), D)$$

Method 2: put boundary emphasis directly into BCE

Boundary-weighted BCE

$$\mathcal{L}_{WBCE} = \frac{\sum_i w_i \text{BCE}(z_i, y_i)}{\sum_i w_i}$$
$$w_i = 1 + 20 \exp\left(-\frac{|D_i|}{0.12}\right)$$

$$\mathcal{L}_{final} = \mathcal{L}_{WBCE} + \mathcal{L}_{Dice}$$

Dice preserves global region overlap; weighted BCE changes which pixel errors dominate the gradient.

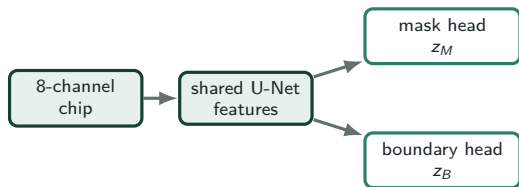
Pixel weight by distance

$ D $	BCE weight
0	21.0
0.12	8.36
0.24	3.71
far from border	≈ 1.0

Key difference

The boundary signal acts on the **final mask output**, not through an auxiliary task.

Method 3: a second head predicts boundary pixels



Why add this head?

Boundary-weighted BCE focuses the mask errors; the second head explicitly teaches the shared features which pixels are borders.

Two-head training objective

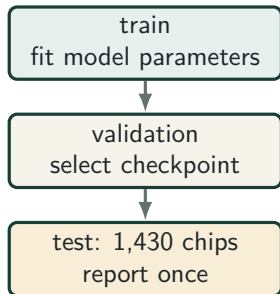
$$B_i = 1[|D_i| \leq 0.12]$$

$$\mathcal{L}_{M+B} = \mathcal{L}_{WBCE}(z_M, Y) + \mathcal{L}_{Dice}(\sigma(z_M), Y) + 2 \mathcal{L}_{BCE}^{bal}(z_B, B)$$

- ▶ $B = 1$: close to a field boundary
- ▶ $B = 0$: all other pixels
- ▶ Class balancing handles the thin positive boundary band
- ▶ Segmentation still comes from the mask head

Two active output heads

Evaluation protocol: model selection and test stay separate



Fair comparison

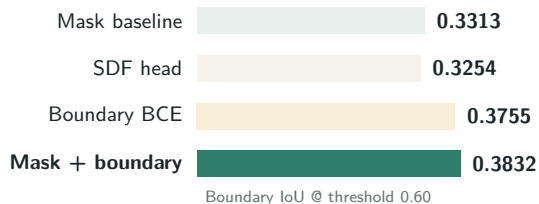
- ▶ Common probability threshold: **0.60**
- ▶ Boundary evaluation tolerance: **2 pixels (≈ 20 m)**
- ▶ Same 1,430 test chips for all models

Metrics

mIoU: region overlap

Boundary IoU: overlap of predicted and target boundary bands

Test result: explicit boundary prediction gives the best BloU



Method	mIoU	BloU
Baseline	.6572	.3313
SDF head	.6500	.3254
Boundary BCE	.6675	.3755
Mask + boundary	.6634	.3832

Main test finding

The explicit boundary head adds **+0.77 pp BloU** over Boundary BCE alone; Boundary BCE alone keeps the highest mIoU.

The boundary head helps; the distance head does not

All four test checkpoints

Test @ 0.60	mIoU	BloU
Baseline	.6572	.3313
SDF head	.6500	.3254
Boundary BCE	.6675	.3755
Mask + boundary	.6634	.3832

What changed

Distance regression

Indirectly modifies shared features; no test gain.

Boundary-weighted mask loss

Directly changes mask gradients; +4.42 pp BloU.

Explicit boundary head

Adds a focused border task; another +0.77 pp BloU.

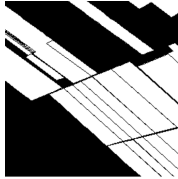
The auxiliary target matters: predicting borders is more useful than regressing distance.

Qualitative comparison on the same Austria chip

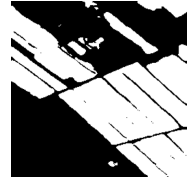
Sentinel-2 RGB



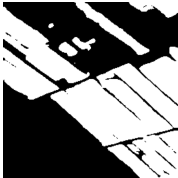
Ground truth



Baseline



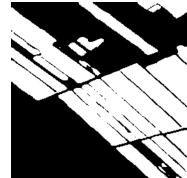
SDF auxiliary head



Boundary BCE



Mask + boundary heads



Same image and probability threshold (0.60) for every prediction.

Source: FTW validation chip `austria/g83_00040_14`; predictions generated in this project.

Conclusion and outlook

Takeaway

Direct boundary supervision is more effective than auxiliary signed-distance regression.

- ▶ Boundary BCE: 0.3313 $\rightarrow$ 0.3755 BIoU (**+4.42 pp**)
- ▶ Adding the boundary head reaches **0.3832 BIoU** (**+5.19 pp**)
- ▶ New model mIoU is 0.6634, still +0.62 pp above baseline
- ▶ The SDF regression head does not improve the test checkpoint

Best boundary model: boundary-weighted mask loss + explicit boundary head.

Next steps

- ▶ repeat with multiple random seeds
- ▶ report per-country performance
- ▶ test instance-separation post-processing
- ▶ study learned boundary calibration

Backup: threshold sensitivity on the test set

Method	0.40		0.50		0.60		0.70	
	mIoU	BloU	mIoU	BloU	mIoU	BloU	mIoU	BloU
Baseline	.6593	.3049	.6587	.3199	.6572	.3313	.6506	.3359
2-head SDF	.6554	.3017	.6528	.3153	.6500	.3254	.6369	.3246
Boundary BCE	.6706	.3364	.6731	.3618	.6675	.3755	.6520	.3682
Mask + boundary	.6717	.3499	.6685	.3705	.6634	.3832	.6467	.3715

Why the main talk uses 0.60

It is a single common operating point for all four models. The new model has the best Boundary IoU at every tested threshold.

Backup: exact objectives

Boundary BCE

$$\mathcal{L}_{BBCE} = \mathcal{L}_{WBCE}(z_M, Y) + \mathcal{L}_{Dice}$$

One mask output; $\alpha = 20$ and $\sigma = 0.12$ define the pixel weights.

Mask + boundary heads

$$\mathcal{L}_{M+B} = \mathcal{L}_{WBCE} + \mathcal{L}_{Dice} + 2\mathcal{L}_{BCE}^{bal}(z_B, B)$$

Two active outputs; the explicit boundary term has weight 2.

How the boundary target is formed

$B_i = \mathbf{1}[|D_i| \leq 0.12]$. The precomputed distance target constructs the weights and boundary labels; the final model predicts only the mask and boundary maps.

Backup: references

-  H. Kerner et al.
Fields of The World: A Machine Learning Benchmark Dataset for Global Agricultural Field Boundary Segmentation.
AAAI, 2025.
-  O. Ronneberger, P. Fischer, and T. Brox.
U-Net: Convolutional Networks for Biomedical Image Segmentation.
MICCAI, 2015.
-  H. Kervadec et al.
Boundary Loss for Highly Unbalanced Segmentation.
MIDL, 2019; extended in Medical Image Analysis, 2021.

All prediction panels and project diagrams were generated from this repository; no external figures are reproduced.